A pioneer in computing and artificial intelligence, mathematician and logician Alan Turing developed code-breaking machines that helped to turn the tide of World War II. He wrote the blueprint for how computing machines could be programmed to perform any task, heralding the advent of the computer age.

MILESTONES

THE TURING MACHINE
Outlines a "universal computing machine" in his 1936 paper "On Computable Numbers."

CRACKING CODES
Plays an instrumental role from 1938 in mathematical and computational methods of code-breaking.

EARLY PROGRAMMING
Compiles his Abbreviated Code Instructions in 1947, the first-ever computer programming language.

THE TURING TEST
Writes a test to assess artificial intelligence in "Computing Machinery and Intelligence" in 1950.

COMPUTING FIRST
Programs the first commercially marketed digital electronic computer in 1951.

Born in London in 1912, Turing was described as a genius by his headmistress at the age of 9, and demonstrated an early interest in mathematics and complex chess problems. He read mathematics at King's College, Cambridge, gaining a undergraduate degree in 1934 and being elected Fellow in 1935 for a thesis on probability theory. In 1936, he turned to the "decision problem," an unresolved question of logic posed by David Hilbert. Turing succeeded in solving the problem, although he was narrowly beaten to it by US logician Alonzo Church. However, the important thing was Turing's method: he solved the problem using a thought experiment about a machine that could be programmed to perform any task—a computer.

The universal computing machine

Turing imagined a machine that could read and write numbers and symbols printed on a long strip of paper, then perform a predetermined task according to an algorithm—a set of rules.

> **"Machines** take me by **surprise with great frequency."**
>
> **Alan Turing**, 1950

***At Bletchley Park**, Turing was the scientific genius behind the vital work of decrypting German codes. The extent of his role was kept secret until the 1970s, and only revealed in full 50 years after his death.*